

Build Advanced Agent with n8n



Instructor: Lekshmi Pillai

Introduction – Lekshmi Pillai

Tech and People Leadership – FinTech, Amazon, Cisco, Accenture, Infosys

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Welcome to today's class, before we begin, pop into the chat:



Structure of Class

1. **Sunday Class - 4 hr (Sunday)**
2. **Technical Coaching Session (Wednesday)**



Wednesday Sessions

Wednesday - Technical Coaching Session

One hour of topical review on the class. This is an excellent time to reinforce Sunday's content or ask advanced questions, depending on your experience.



Optimize Your Experience

- ✓ Interact with your instructors via Sunday live class.
- ✓ Don't be shy to speak up and get clarifications !



- ✓ Solve your assignments, MCQs and other assessments and get feedback
- ✓ Use your resources! It's your experience – what you put in is what you'll get out.

Don't worry!

We are here to help at your every step !

But, how???



Are you using all these IK support features?

- **Uplevel** - contains all the learning resources - videos, MCQs, assignments
- Optional **Post class videos** for additional reference
- **Technical coaching sessions** on Wednesday to clarify all your queries
- **Discord** community for peer learning
- **TAs** to help answer the queries in the live class
- **Support tickets** - Raise them any time you want any clarifications/tech concerns , etc on UpLevel

Success Hacks

- Be prepared for the class by watching the pre class videos
- Attend every class;
- Participate in class
- Attempt all the assignments,
- Actively advocate for yourself
- Keep your goal in mind
- Consistency is the key

Be patient with yourself !

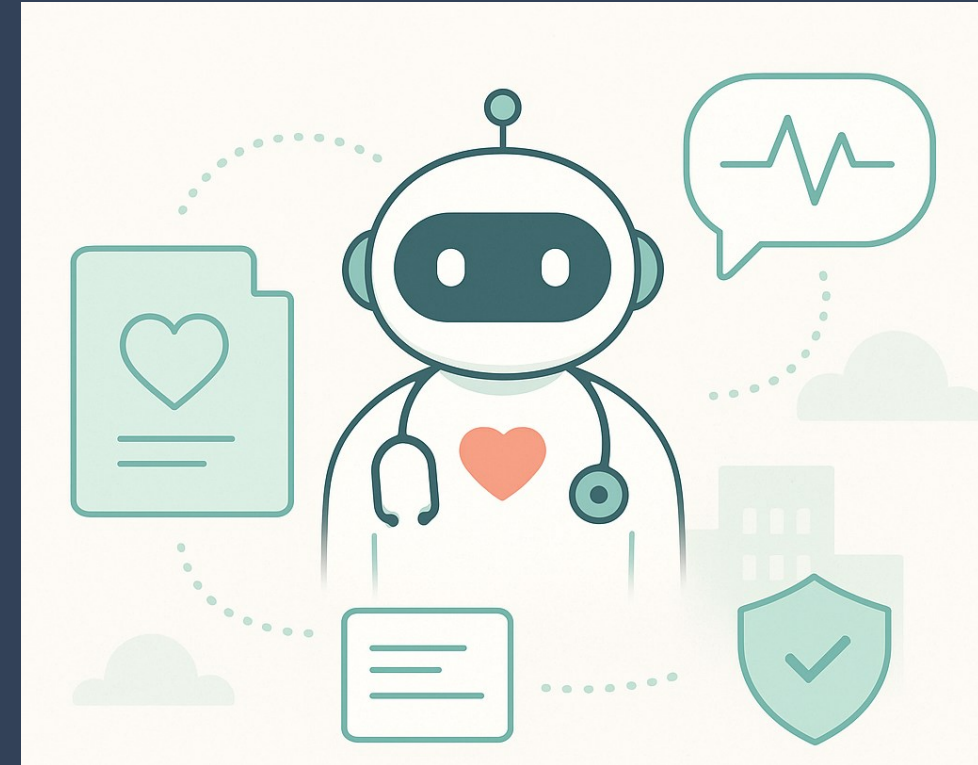
Session Goals

- Build an advanced, real-world multi-agent system on n8n.
- Learn to orchestrate specialized agents using structured healthcare data.
- Practice scaling, testing, and debugging multi-agent workflows.

**Let's understand the problem statement that
lies in front of us in detail.**

Problem Statement

- Accessing reliable and personalized health information remains a challenge for many individuals.
- Traditional healthcare information systems provide generic responses, often failing to consider individual medical history or maintain conversational context.
- Let's build HealthSense AI, an intelligent, Multi- Agent healthcare system using n8n to provide personalized, context-aware health insights while ensuring medical accuracy and compliance.



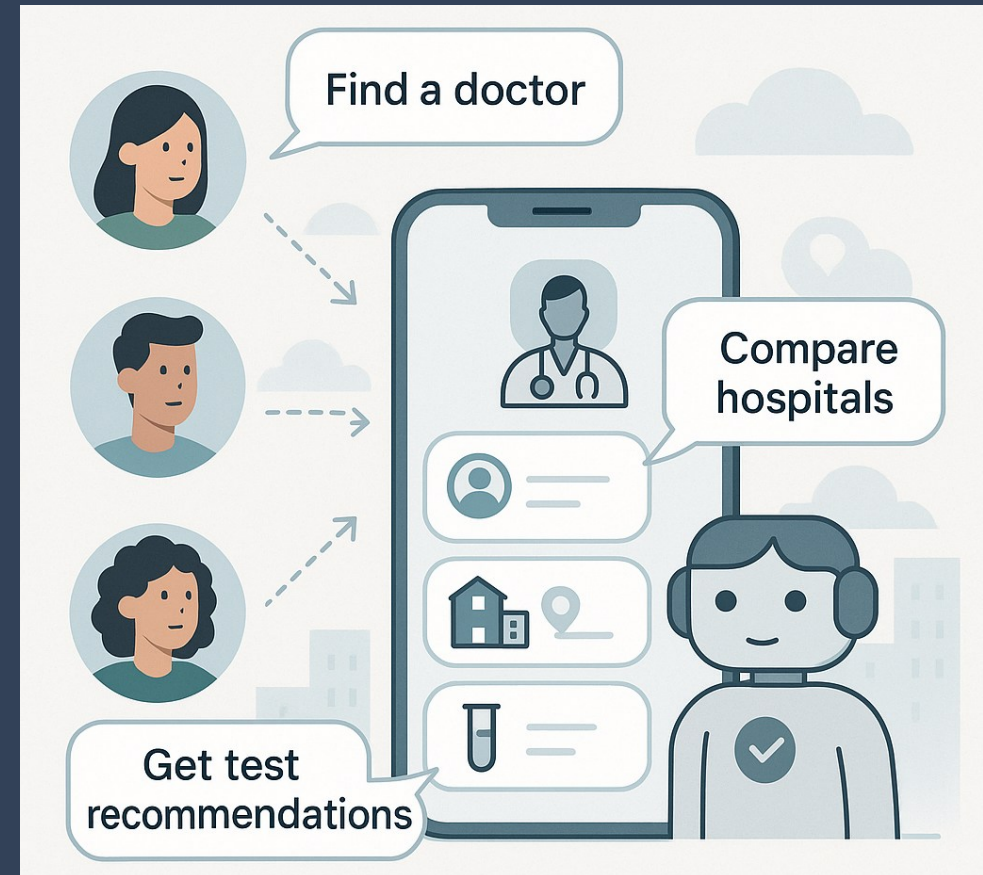
Problem Statement

Users want:

- To Find available doctors and book appointments.
- To compare hospitals based on specialization, services, and emergency availability.
- To get diagnostic test recommendations.

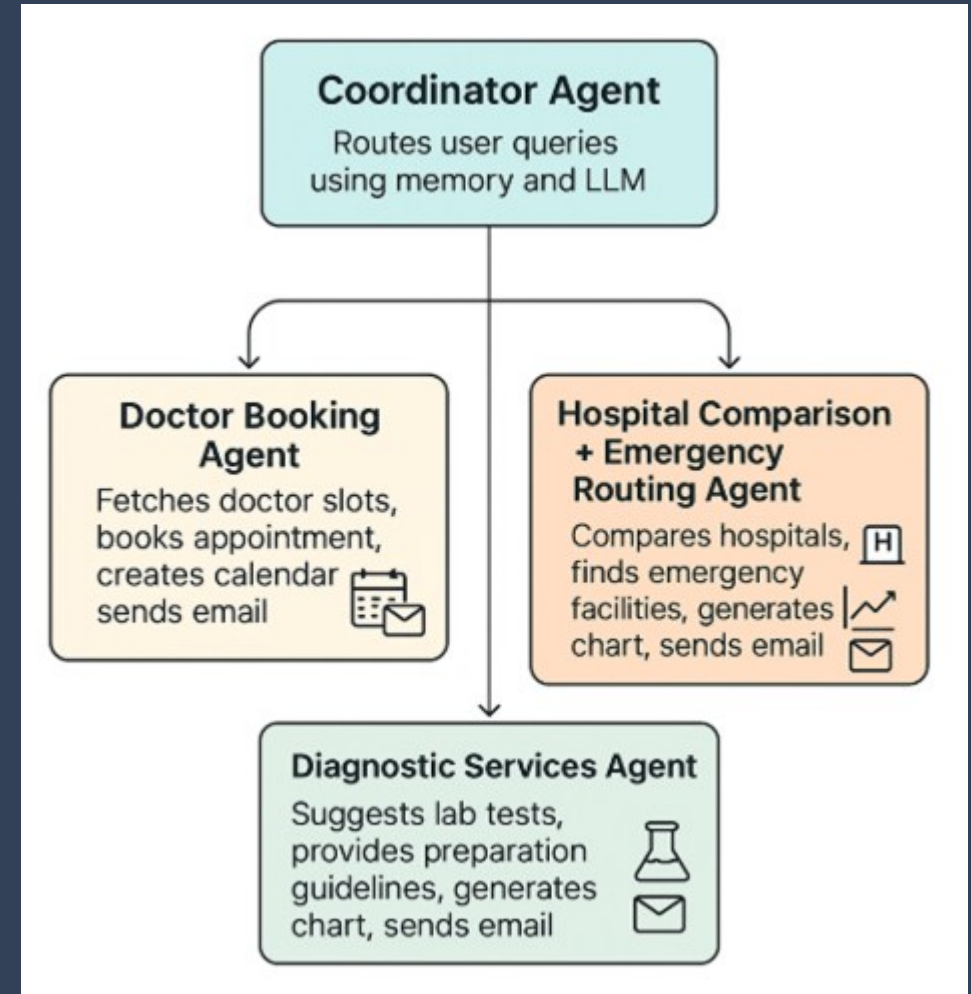
Example User Queries:

- "Find an available dermatologist this week."
- "Which hospitals near me have cardiology departments?"
- "What tests should I take for persistent



What You Will Build Today

- **A modular, advanced multi-agent system that can:**
 - Book doctor appointments
 - Compare hospitals and find emergency services
 - Retrieve diagnostic test recommendations
- Fully orchestrated using a **Coordinator Agent** on n8n.
 - To route user queries to specialized healthcare agents intelligently.



Understanding Your Dataset

What Data Are We Using?

You will work with **structured healthcare datasets** in Google Sheets to power your agents:

Dataset	Fields	Description
Doctors Info	Name, Specialization, Contact Details	Basic doctor profiles including specialty and contact info
Doctors Slots	Doctor Name, Available Appointment Times	Available slots linked to each doctor
Hospital Information	Hospital Name, Specialties, Address, Ratings	Hospital profiles with services offered and ratings
Emergency Data	Hospital Name, Ambulance Availability, Emergency Availability	List of hospitals with ambulance and ER readiness
Lab Tests Information	Test Name, Preparation Instructions	Diagnostic test details with preparation guidelines

n8n - Recap



- **Open-source** automation tool for building custom workflows and integrations.
- **Connects to various apps, services, LLMs, and APIs** (e.g., Google Sheets, GPT, AirTable, Slack, CRMs, databases).
- **No-code/low-code interface** for building workflows and automating tasks without coding.
- Enables **building and managing agents** to perform specific actions autonomously across services.
- Self-hosted or cloud options, giving **flexibility, scalability, and control** over automation and agent management.

Let's start building our agents!

1. Coordinator Agent

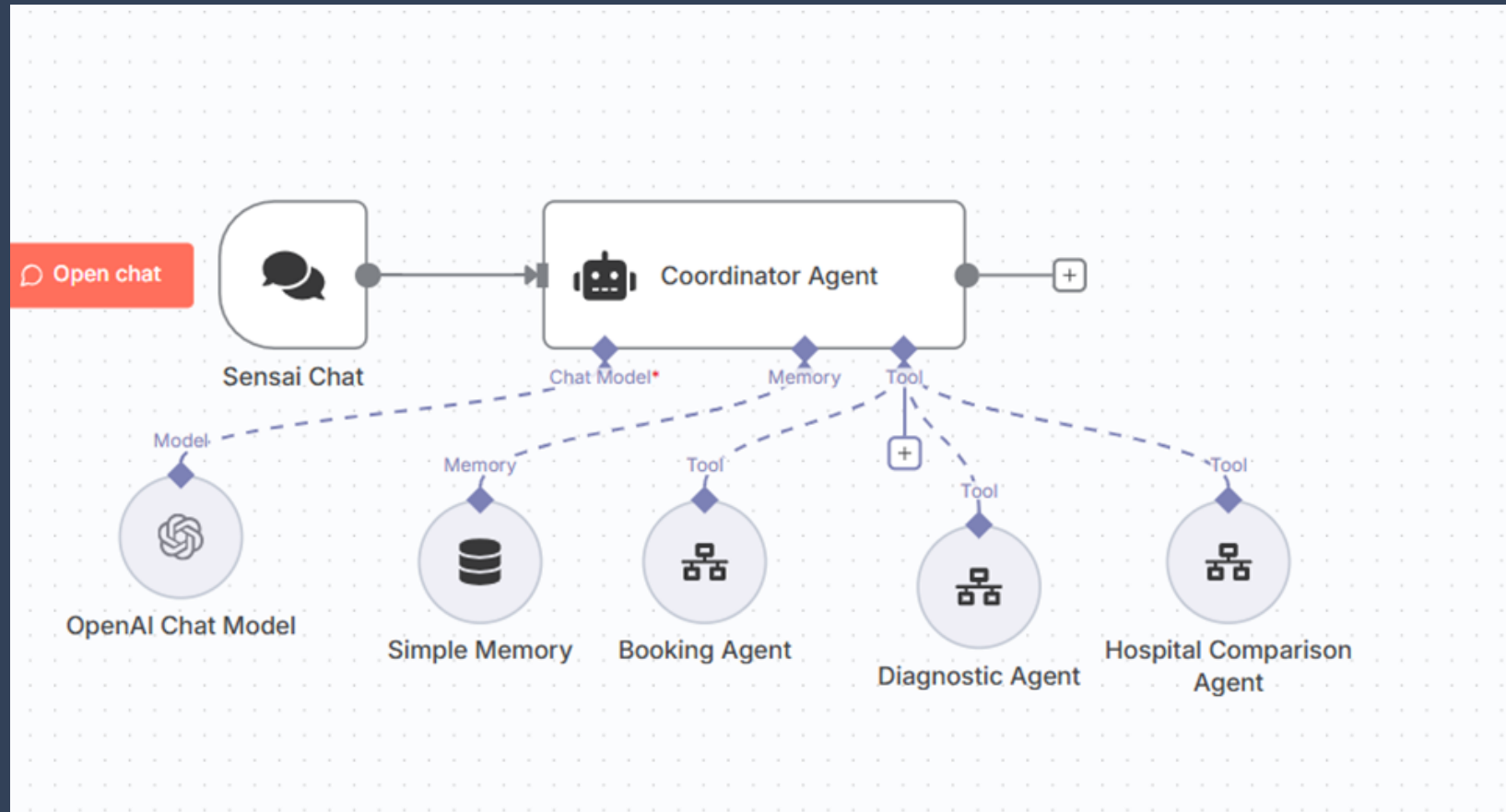
Purpose:

- Acts as the **brain of your system**, routing user queries to the correct specialized agent automatically.

Functionality:

- Receives user queries via chat trigger.
- Uses **OpenAI** for intent understanding.
- Uses **Memory Buffer** for maintaining context.
- Calls the **Booking Agent, Hospital Agent, or Diagnostics Agent** as needed using **ToolWorkflow**.

1. Coordinator Agent



1. Coordinator Agent

Sample User Queries:

- "Find a dermatologist available this week."
- "Show me hospitals with cardiology departments."

Output:

User queries are routed accurately to the appropriate specialized agent for processing.

n8n Demo

2. Doctor Booking Agent

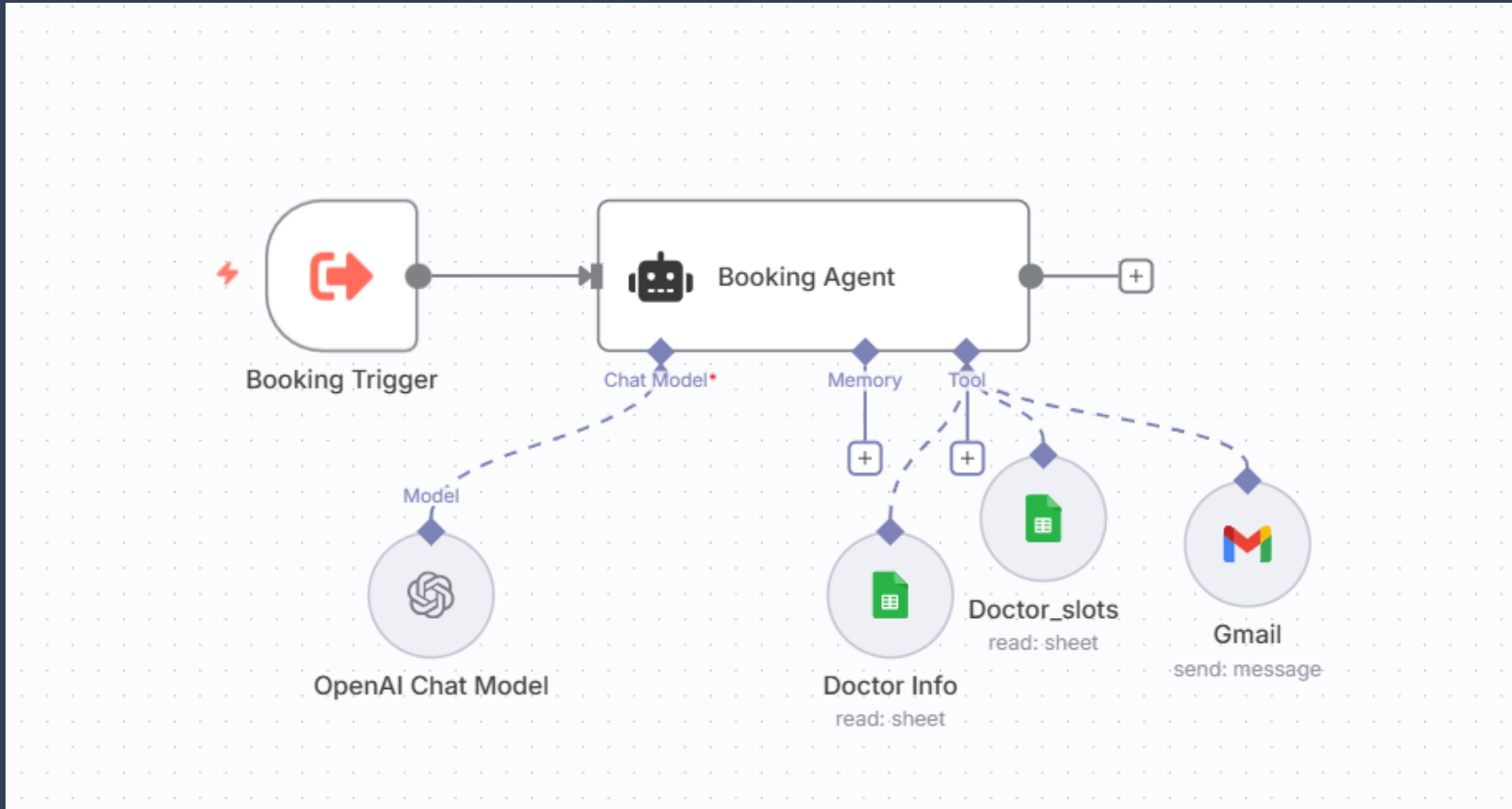
Purpose:

- Handles **doctor availability and appointment booking** seamlessly within your system.

Functionality:

- Filters available doctors based on specialty or name.
- Retrieves available slots for appointments.
- Confirms and sends email booking confirmations.

2. Doctor Booking Agent



2. Doctor Booking Agent

Key Nodes/Components:

- **Google Sheets:** Doctor Availability and Doctor Info datasets.
- **OpenAI Chat:** For interpreting user input.
- **Function Node:** To filter and format slots.
- **Gmail:** To send confirmation emails.

Sample User Queries:

- "Find an available dermatologist."
- "Show me slots for Dr. Lee next Monday."

Output:

List of available doctors and slots, with an email confirmation sent upon booking.

n8n Demo

3. Hospital Comparison + Emergency Routing Agent

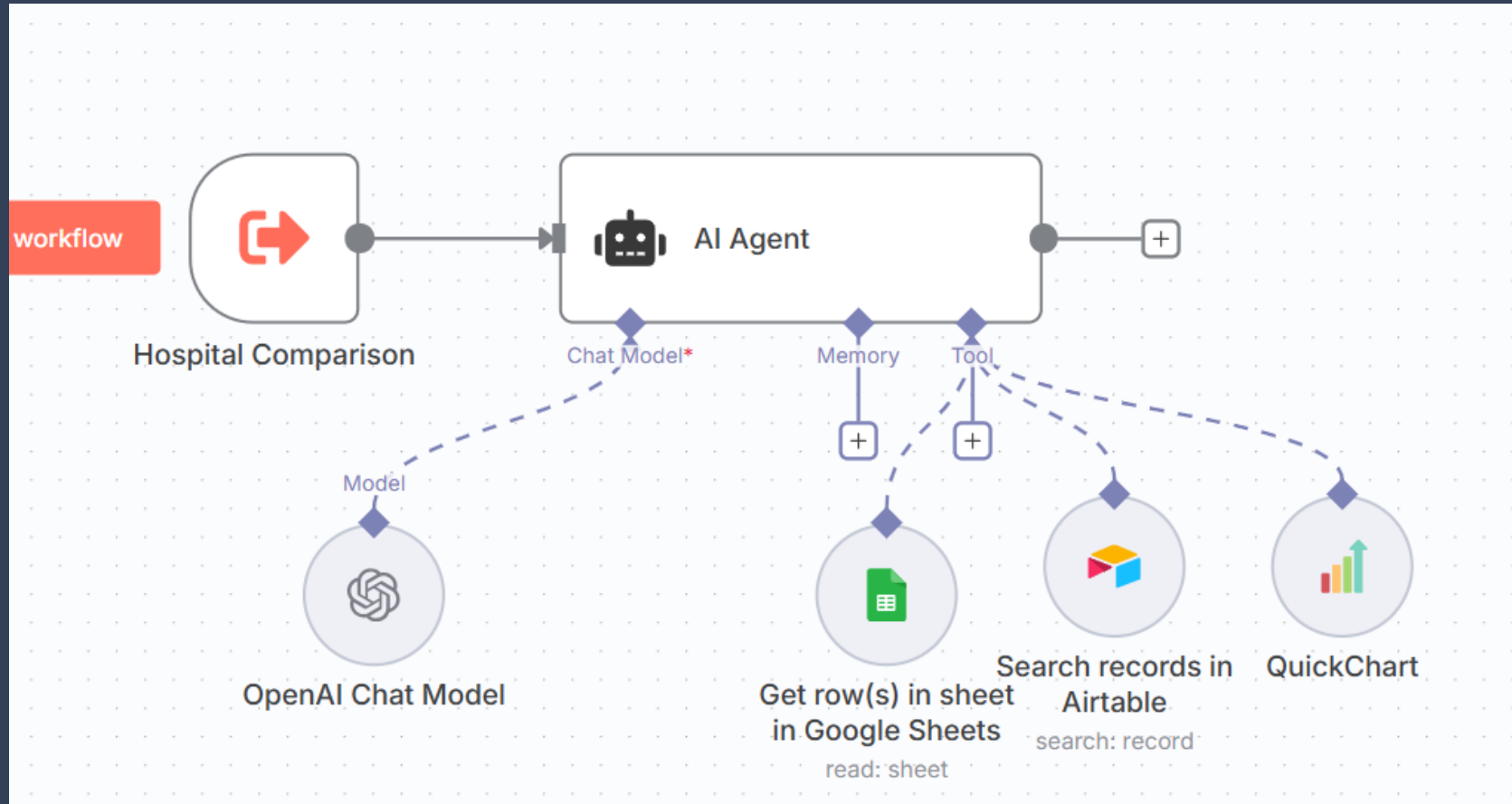
Purpose:

- Helps users **compare hospitals and find emergency services** efficiently.

Functionality:

- Compares hospitals by specialty, ratings, and proximity.
- Finds the nearest emergency hospitals.
- Visualizes hospital rankings using charts.

3. Hospital Comparison + Emergency Routing Agent



3. Hospital Comparison + Emergency Routing Agent

Key Nodes/Components:

- **Google Sheets:** Hospital General Info and Emergency Data.
- **Function Node:** For proximity and emergency calculations.
- **OpenAI Chat:** For formatting comparison output.
- **QuickChart:** To display hospital ratings visually.

Sample User Queries:

- "What hospitals specialize in cardiology near me?"
- "Where is the nearest 24/7 emergency hospital?"

Output:

Provides a sorted hospital list, nearest emergency services, and visual charts of hospital rankings.

n8n Demo

4. Diagnostic Services Agent

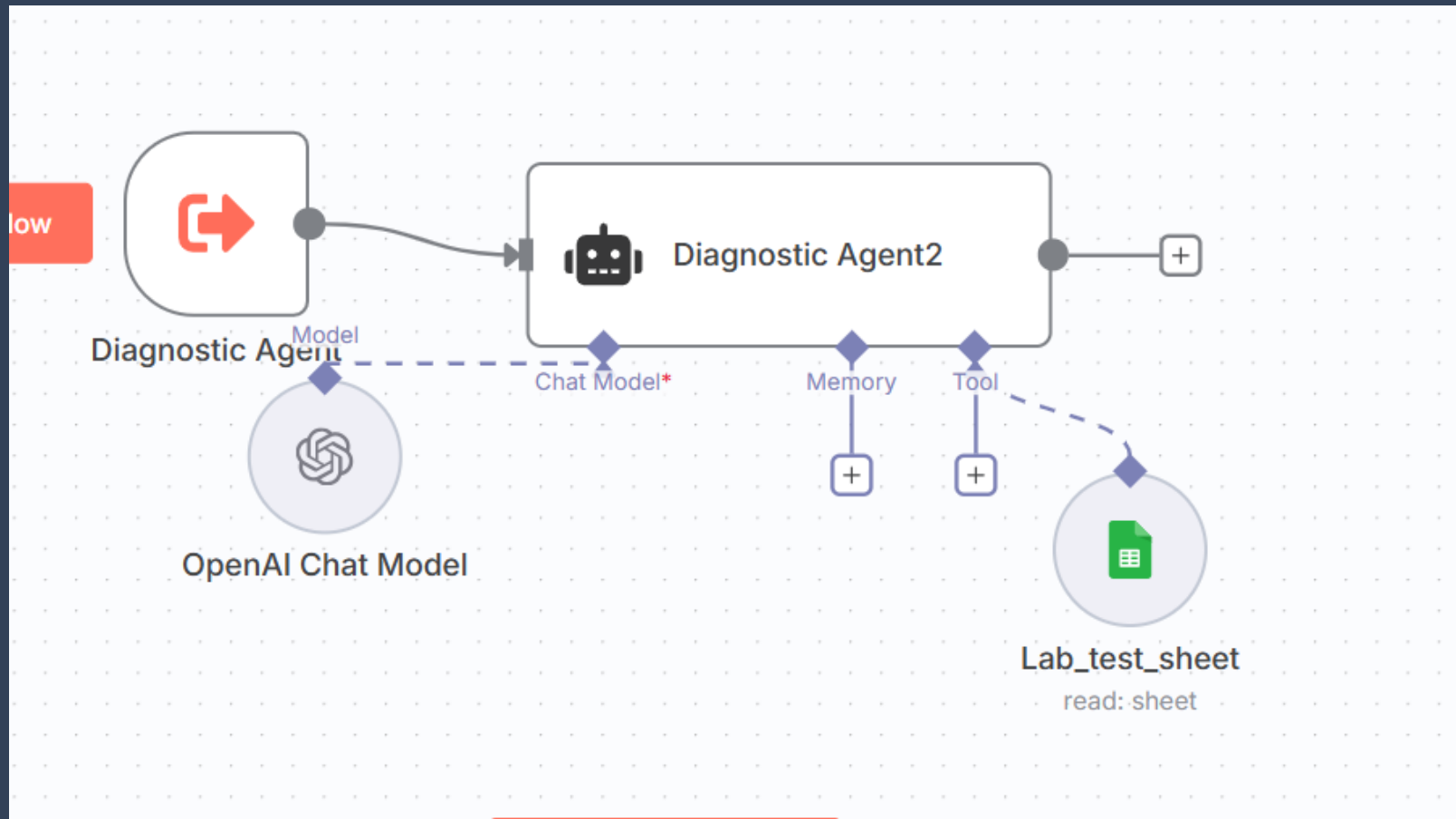
Purpose:

- Helps users **get lab test recommendations and preparation guidelines.**

Functionality:

- Suggests lab tests based on user symptoms.
- Provides detailed preparation instructions.
- Optionally sends preparation instructions via email.

4. Diagnostic Services Agent



4. Diagnostic Services Agent

Key Nodes/Components:

- **Google Sheets:** Lab Tests and Preparation dataset.
- **OpenAI Chat:** For mapping symptoms to tests.
- **Gmail:** For sending preparation guides.

Sample User Queries:

- "What tests should I take for persistent headaches?"
- "What are the preparation instructions for a cancer screening?"

Output:

A list of recommended tests with preparation instructions, with optional email delivery for user convenience.

n8n Demo

Take Home exercise: Extending & Scaling System

Add a Pharmacy Availability Agent to your system.

Steps to Guide Your Experiment:

- Create a **dummy dataset in Google Sheets** with columns: Pharmacy Name, Address, City, State, Zip Code, Open 24/7 (Yes/No), Contact Number. Add 10 sample entries manually.
- Create a new n8n workflow to handle pharmacy queries.
- Connect the Google Sheets tool to read your pharmacy data.
- Prompt the agent to respond to queries like “Find 24/7 pharmacies near 10001” or “List pharmacies in San Francisco.”
- Test end-to-end by calling this new agent from your Coordinator Agent.

**Thank
you**